

# Miles V. Vollmer

Curriculum Vitae: Software & Data Platform Engineering

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## PROFESSIONAL SUMMARY

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Performance and data-platform engineer with a Computer Science / AI background from Northeastern University, currently consulting for enterprise clients through Moviri. Builds and ships production ETL pipelines, BI platform migrations, and full-stack automation across banking and telecommunications environments. Recent work spans reverse-engineering undocumented BI platform internals, translating a 451-dashboard legacy Grafana estate into Databricks AI/BI, and leading development of an award-winning enterprise dashboard for a major UK bank. Comfortable operating independently across the full stack: source-system SQL, backend services in Java/Python, and the API and platform layers that connect them.

## EDUCATION & CERTIFICATIONS

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**B.S. in Computer Science, AI Concentration**, Northeastern University, Boston, MA, 2024

**Certifications:** AWS Cloud Practitioner; BMC Certified Associate

## TECHNICAL SKILLS

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**Cloud & Data Platforms:** Databricks (Lakeview / AI-BI dashboards, Unity Catalog, SQL Warehouses, Lakehouse Federation), AWS (EC2, S3), Grafana, Tableau, Power BI, dbt, BMC Helix (Continuous Optimization), Dynatrace, Docker, Kubernetes, CI/CD, Git, NetSuite, Bash scripting

**Programming & Development:** Python, Java, SQL, R, JavaScript, TypeScript, HTML, CSS

**Automation & Platform Engineering:** ETL pipeline design, BI platform migration, SQL transpilation (Redshift/PostgreSQL → Databricks SQL), REST API integration and reverse-engineering, agentic and multi-agent automation, release/CI-CD automation, system monitoring, process optimization

**Machine Learning & AI:** PyTorch, TensorFlow, NLP, HuggingFace, Google ADK, multi-agent systems, reinforcement learning, SHAP-based feature analysis

**Other:** LLM and RAG systems development, prompt engineering, full-stack development, web development and API usage, agentic systems, technical documentation for cross-functional and non-technical audiences

## PROFESSIONAL EXPERIENCE

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### Performance Engineering Consultant, Moviri

Jan 2025 – Present

Consulting engineer embedded with two enterprise clients, leading platform migration, BI modernization, and AI-driven automation workstreams end to end, from source-system analysis through production delivery.

### Lloyds Banking Group: BMC Helix Continuous Optimization (BHCO)

- Led front-end and back-end development of a Business Units dashboard for BHCO with multi-level drill-down, letting users navigate via simple clicks from the highest platform view down to the root cause for any business unit. The dashboard won the **Innovation Award** at the London BMC Helix Roadshow and was recognized company-wide as the client's standout implementation.
- Migrated client enterprise systems onto the BMC Helix cloud platform; led capacity analysis that right-sized ~10,000 overprovisioned servers, cutting infrastructure spend by \$2M+ per month.
- Extended a Dynatrace ETL collector (Java) to capture GCP host metadata (availability zone, project ID, instance type, resource group) across 2,300+ hosts. Designed a three-tier fallback resolution strategy (host properties → Kubernetes labels → GCP entity API) that avoided an expensive relationship-graph query pattern which would have been impractical at that host count.
- Built a Java HTTP client for the BHCO REST API, wired into the team's existing authentication and request-building infrastructure (shared *RequestExecutor/LoginInfo* pattern) rather than duplicating it, keeping the new extractor consistent with the rest of the codebase.
- Diagnosed and resolved dashboard configuration issues, including a threshold-configuration bug tied to a metrics API call and a history-chart time-filter fix; added saturation-risk indicators (total/minimum memory and CPU, days-to-saturation) to two production dashboards.

- Authored technical construction documentation for four reusable ETL extractor modules (Dashboard Dependencies, Datamart Lineage, ETL Primary Datamart, License Count) and their corresponding dashboard integrations, written for readability across technical and non-technical stakeholders.

#### **Fastweb: Grafana-to-Databricks BI Migration**

- Architected and built a ~15-tool Python toolchain migrating a legacy Grafana BI estate (**451 dashboards, 4,150 panels**, PostgreSQL/Redshift-backed) into Databricks AI/BI (Lakeview). Toolchain covers inventory and migration-complexity scoring, SQL transpilation, widget-prototype extraction, spec-driven JSON dashboard assembly, numeric validation against source baselines, and governed delivery into a separate production repo.
- Migrated over 400 dashboards into the target repository, including network-operations dashboards for packet-loss/RTT/jitter monitoring and optical transport capacity planning; matching the source system's UX (multi-series stat panels, threshold-based coloring, cascading filters) within a different BI platform's constraints, through multiple stakeholder review cycles.
- Reverse-engineered undocumented Databricks Lakeview internals with no public schema: dynamic widget titles bound to live filter state, per-widget typography controls, legend visibility/positioning, conditional per-cell formatting, and color-fill mechanics; each verified empirically against the live product and turned into a reusable, tested build capability rather than a one-off hack.
- Root-caused data-correctness issues rather than accepting cosmetic explanations; for example, traced an apparent 0.03-percentage-point discrepancy between platforms to a live rolling-window boundary effect on a specific spike hour, by querying source warehouses directly.
- Translated Redshift/PostgreSQL SQL to Databricks SQL at scale: resolved timezone-conversion bugs, translated Grafana template variables into SQL parameters, and solved a *WITH RECURSIVE* + Lakehouse Federation incompatibility via an unrolled-join rewrite.
- Designed process guardrails for AI-agent-assisted engineering at scale: a dated decision log preventing rework across concurrent sessions, dry-run-first delivery tooling, and a 52-test automated regression suite protecting the build pipeline.
- Authored client-facing technical documentation translating platform capability gaps into plain-language design rationale for non-technical stakeholder conversations.

#### **Additional Automation Delivery (Moviri, cross-client)**

- Designed and implemented an automated YAML-based CI/CD release pipeline for monthly product releases across all offerings, reducing manual deployment intervention.
- Built a full-stack automated HR system for meal voucher allocation, integrating backend logic with a user-facing UI to eliminate hours of monthly manual work.
- Developed a multi-agent-powered onboarding automation system, integrated with NetSuite, that gathers and updates new-hire data, accelerating onboarding timelines and reducing HR workload.
- Designed and built Human-Oriented Automation Orchestration (HOAO), an internal multi-agent orchestration framework (Google ADK/Gemini) across six build iterations, with a human-in-the-loop pipeline where agents draft, sandbox-test, and self-validate their own automation tools before caching them for reuse; piloted internally as a deployed Cloud Run tool under Moviri's AI Labs program.

#### **Research Assistant, Evolutionary Genomics Lab, Northeastern University**

May 2021 – Sep 2021, Sep 2022 – Aug 2024

- Developed multiple machine learning classifiers (Multilayer Perceptron, Support Vector Machine, Random Forest, Lasso Regression, and others) to predict coral disease states from 16S rDNA bacterial gene sequencing data, achieving over 97% classification accuracy.
- Conducted feature-importance analysis using SHAP values to identify the bacterial taxa most predictive of disease state.
- Co-authored a peer-reviewed publication in *Environmental Microbiology* detailing the machine learning-based pathogen identification pipeline and findings on top candidate pathogens (see Publications).

#### **Data Research Analyst, Iafate Lab, Massachusetts General Hospital (MGH)**

Jan 2022 – Aug 2022

- Developed an automated pipeline for Genome-Wide Association (GWA) analysis of COVID-19 data, improving processing efficiency for large-scale genetic analysis.
- Analyzed demographic variation in positivity rates using advanced data visualization, producing insights that informed healthcare intervention and policy discussions.

## **PUBLICATIONS**

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Selwyn, JD, Despard, BA, Vollmer, MV, Trytten EC, Vollmer, SV (2024). "Machine Learning based identification of putative coral pathogens in endangered Caribbean staghorn coral." *Environmental Microbiology* 26 (9): e16700. [enviromicro-journals.onlinelibrary.wiley.com/doi/epdf/10.1111/1462-2920.16700](https://onlinelibrary.wiley.com/doi/epdf/10.1111/1462-2920.16700)

## SELECTED PROJECTS

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**Coral Population Genomics Pipeline:** Two-pass Snakemake/ANGSD population-genomics pipeline across 290 whole genomes (2 *Acropora* species, 3 Caribbean regions), portable across workstation, SLURM HPC, and AWS EC2 (Singularity). Retained 2,034,805 SNPs; FST analysis quantified near-complete species isolation (0.43–0.46 inter-species vs. 0.05–0.11 intra-species), SMC++ reconstructed demographic history for both species, and LFMM genotype-environment association identified a candidate thermal-tolerance locus (RNF213) corroborated by an independent introgression signal. Diagnosed and fixed a recombination-mapping (pyrho) bug blocking downstream analysis. Engineered an AWS spot-instance cost plan (~70% savings vs. on-demand) with automated retry, S3 sync, and auto-stop. Feeds an in-preparation publication.

**No-Limit Texas Hold'em Poker Agent:** Implemented and benchmarked competing 6-player poker agents (CFR, MCCFR, a custom Q-learning agent, and a REBEL-style self-play agent with neural policy/value networks) via head-to-head tournaments.

**4-Player Chess AI:** Minimax engine with move ordering, alpha-beta pruning, and global search caching for a 2v2 team chess variant on a 14x14 board; 100% win rate vs. random AI, ~50% vs. experienced human players.

**Extracting Traffic Data with NLP:** Built a pipeline over Boston 311 data to detect and classify traffic violations for urban planning use.

**Image Processing System:** Java application supporting advanced filters and format conversions.

## OTHER INTERESTS

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Soccer (Co-Captain of Northeastern FC in the Boston Metro Sunday Soccer League), Surfing, Scuba Diving, Chess, Traveling, Hiking